Gold Price Forecasting Using Machine Learning

Course: FA-582 – Foundations of Financial Data Science

Semester: Fall 2025

Group: Team 5

Team Members:
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Abstract

Gold is widely regarded as a safe-haven asset and plays a critical role in global financial markets. Accurately forecasting gold prices is therefore of significant interest to investors, policymakers, and risk managers. This project investigates the effectiveness of multiple statistical and machine learning models for **one-day-ahead gold price prediction**, using historical gold prices combined with macroeconomic and market indicators from 2011 to 2025.

We evaluate traditional time-series and baseline approaches (naïve persistence and ARIMA), linear regression, and several machine learning models including Random Forest, Gradient Boosting (XGBoost), Decision Trees, and classification-based models for price direction. Our results demonstrate that **Linear Regression and ARIMA outperform more complex machine learning models in out-of-sample forecasting**, primarily due to the strong autocorrelation and trending behavior of gold prices. Tree-based models suffer from severe overfitting and fail to extrapolate beyond the training range.

The findings highlight an important lesson in financial data science: **model complexity does not guarantee superior predictive performance**, particularly in trending time-series data. The project emphasizes interpretability, robustness, and alignment between model assumptions and data characteristics.

1. Introduction

Gold has historically served as a hedge against inflation, currency depreciation, and geopolitical uncertainty. Its price dynamics are influenced by a combination of **macroeconomic indicators**, **financial market conditions**, and **technical price behavior**. With the increasing availability of financial data and advances in machine learning, there is strong interest in applying ML models to forecast gold prices.

However, financial time series pose unique challenges, including non-stationarity, volatility clustering, and structural trends. In this project, we aim to answer the following research questions:

1. Can machine learning models outperform traditional statistical approaches in short-horizon gold price forecasting?
2. How do different model families behave when exposed to strong price trends?
3. Does improved price prediction accuracy translate into profitable trading strategies?

To address these questions, we conduct a systematic comparison of multiple regression and classification models using a consistent experimental framework.

2. Data and Methods

2.1 Data Description

The dataset consists of **3,899 daily observations** spanning **January 2011 to December 2025**, covering approximately **15 years** of market activity. The primary target variable is the **daily closing price of gold (USD per ounce)**.

Predictor variables include:

- Equity market indicators: **S&P 500, Dow Jones Index**
- Commodity prices: **Oil (USO), Platinum, Palladium**
- Currency indicators: **U.S. Dollar Index (USD), EUR/USD**
- Gold-related equities: **GDX**
- Derived technical indicators based on gold prices

Data loading complete! ✓ Successfully loaded data from 2011-2025 covering 14 years.

Figure 1: Historical gold prices (time-series plot)

2.2 Exploratory Data Analysis (EDA)

EDA reveals a **strong upward trend** in gold prices, especially after 2020, along with periods of heightened volatility. Normalized time-series comparisons show that gold prices move inversely with the U.S. Dollar Index and display moderate co-movement with equity and commodity markets.

Correlation analysis confirms that **lagged gold prices and moving averages** are among the most informative predictors, while macro variables provide incremental but weaker explanatory power.

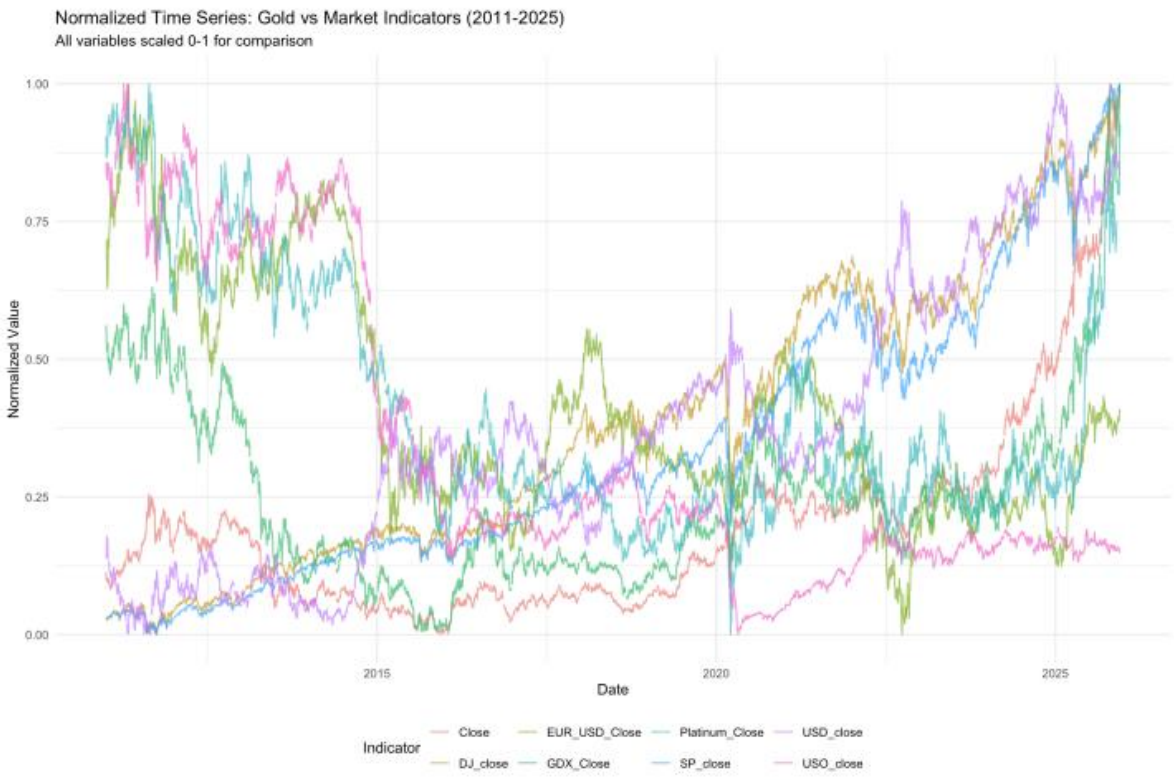

Figure 2: Normalized time-series comparison of gold and market indicators

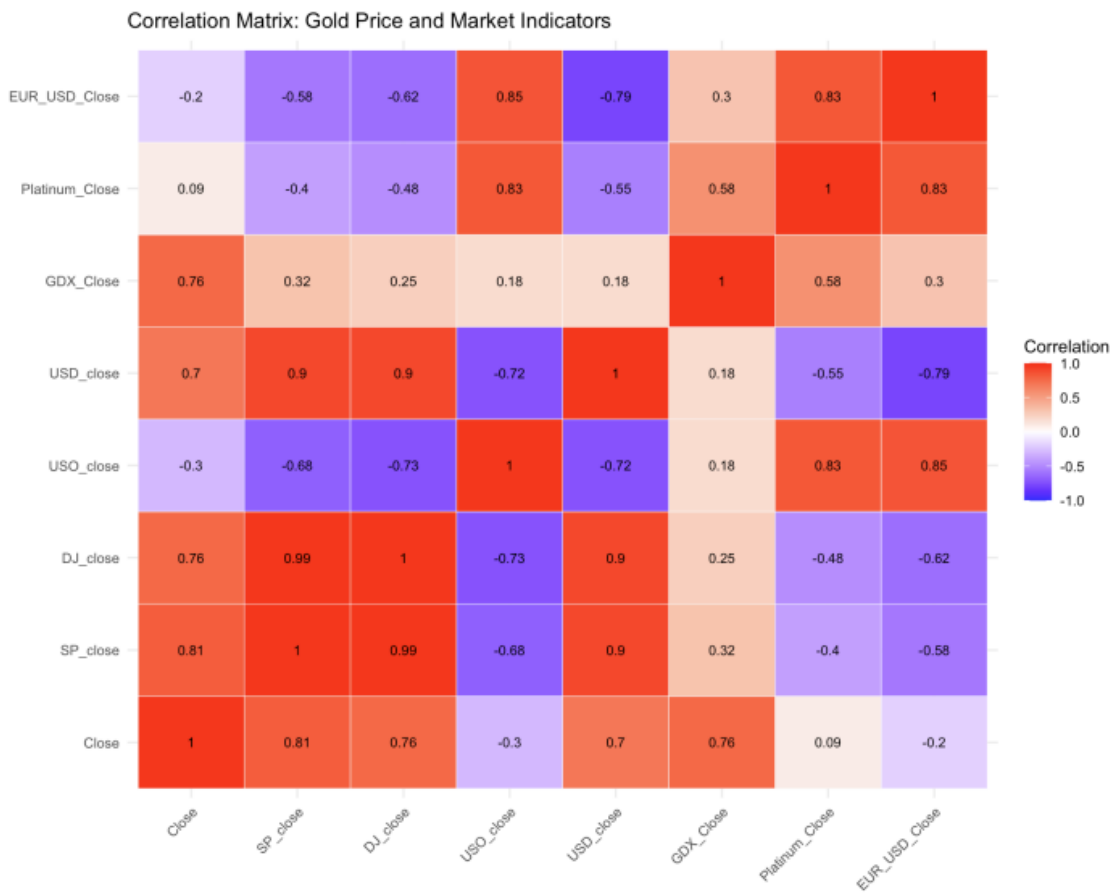

Figure 3: Correlation heatmap of key variables

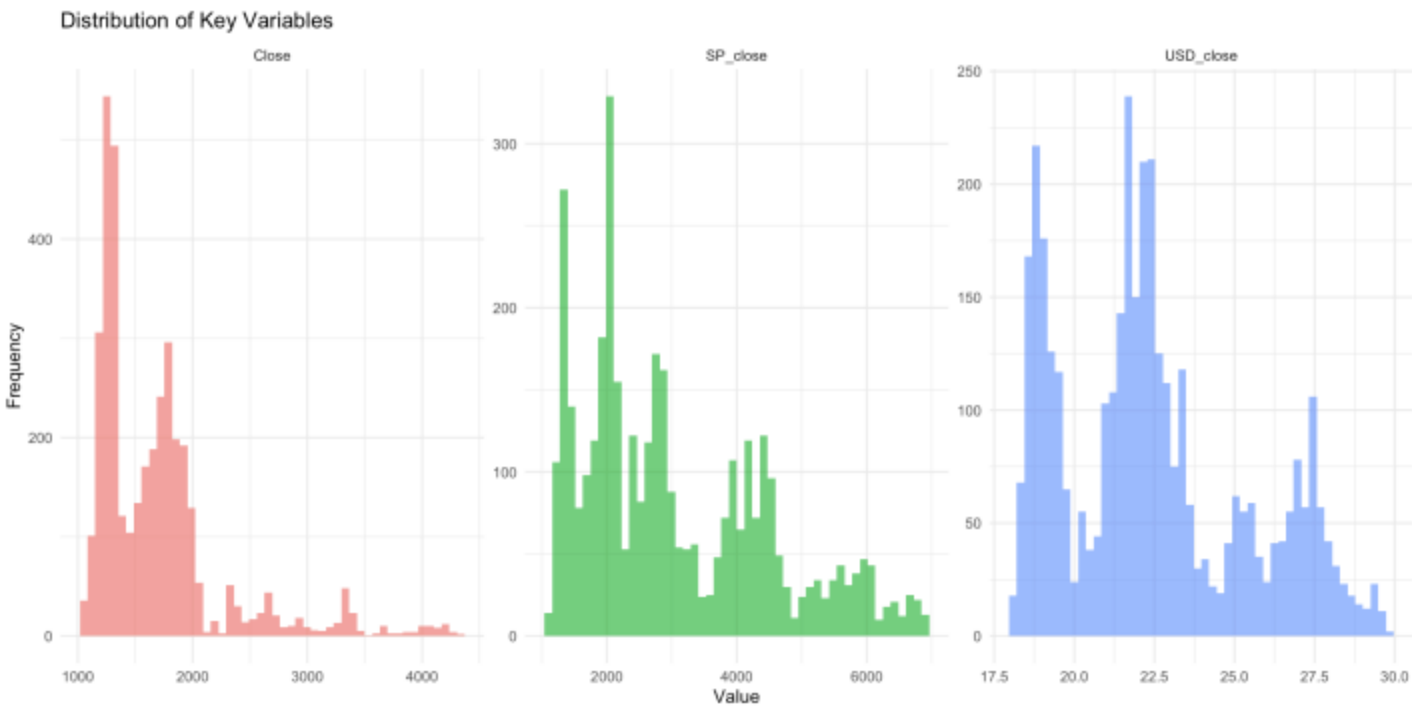

Figure 4: Distribution plots of major predictors

2.3 Data Preprocessing and Feature Engineering

Missing values (typically below 4%) were handled using **forward-fill imputation**, appropriate for financial time-series continuity. Feature engineering included:

- One-day-ahead target: Close_t1
- Lagged prices (1, 3, and 5 days)
- Technical indicators: **7-day and 30-day moving averages**
- Bollinger Band deviation
- Daily returns and log-returns

These features capture both **momentum** and **volatility**, key components of financial price behavior.

Figure 5: Moving averages and Bollinger Bands visualization

2.4 Train–Test Split

The dataset was split chronologically to avoid look-ahead bias:

- **80% training data (3,095 observations)**
- **20% testing data (774 observations)**

This mirrors real-world forecasting conditions and ensures realistic out-of-sample evaluation.

2.5 Models Evaluated

We evaluated the following models:

- **Baselines:**
  - Naïve persistence (previous day's price)
  - ARIMA on log-returns
- **Regression Models:**
  - Linear Regression
  - Random Forest
  - XGBoost
  - Decision Tree
- **Classification Models (Direction Prediction):**
  - K-Nearest Neighbors
  - Multi-Layer Perceptron (Neural Network)

All models were trained using a consistent preprocessing recipe and evaluated using **RMSE, MAE, and R^2** for regression and **accuracy** for classification.

3. Results

3.1 Regression Performance

The **Linear Regression and ARIMA models achieved the best out-of-sample performance**, with RMSE values around **30 USD** and R^2 values close to **0.998**. Surprisingly, these simple models outperformed more complex machine learning approaches.

Tree-based models (Random Forest, XGBoost, Decision Tree) exhibited **extreme overfitting**, achieving near-perfect in-sample performance but failing catastrophically on test data due to their inability to extrapolate upward price trends.

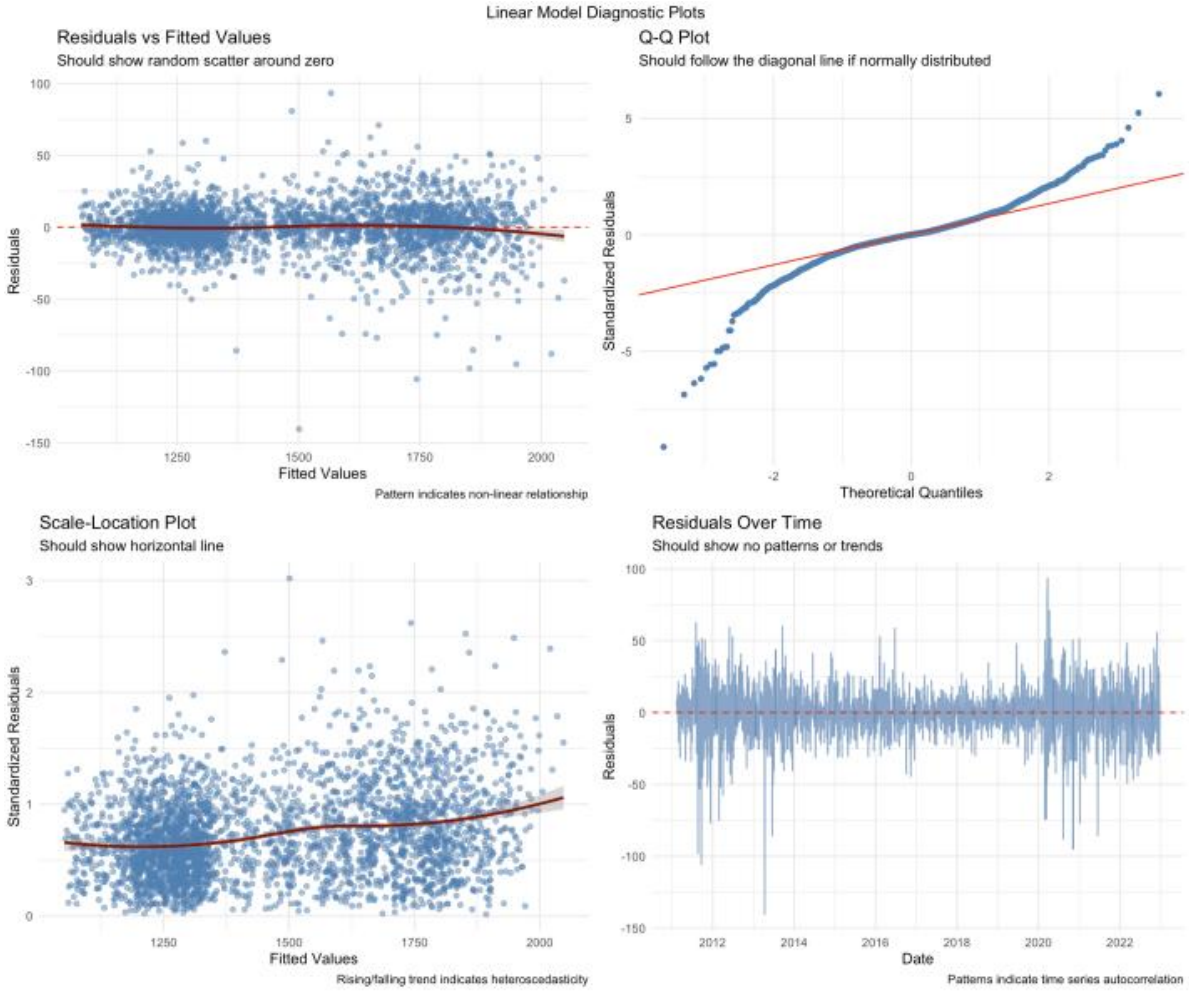

Figure 6: Model predictions vs actual prices (test set)

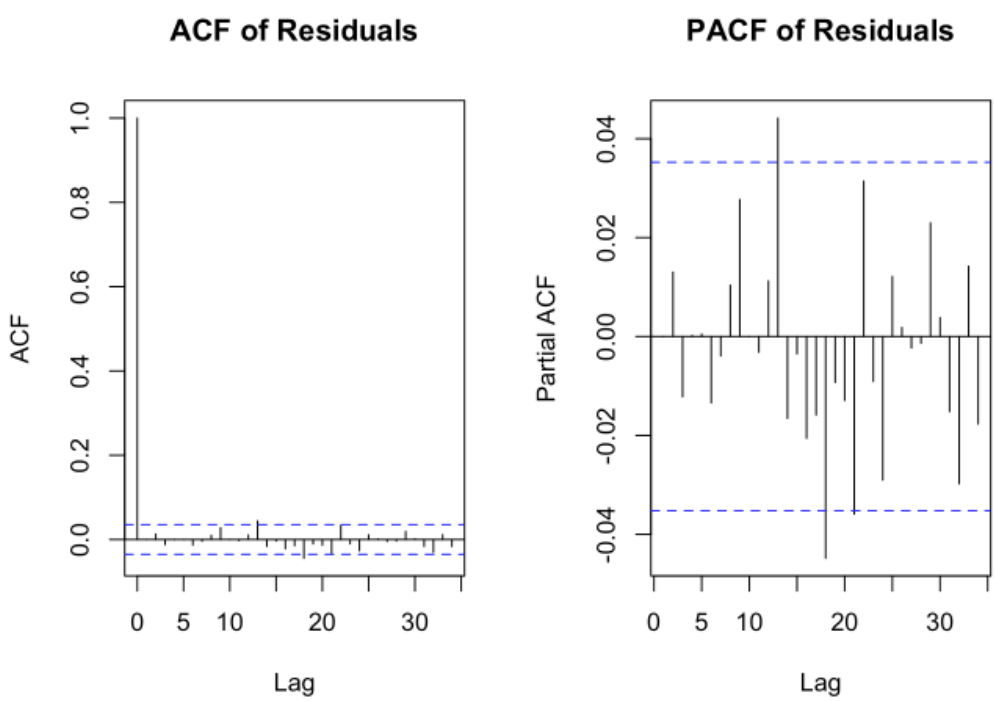

Figure 7: Model performance comparison bar charts

3.2 Overfitting and Model Limitations

Random Forest training RMSE was approximately **7 USD**, while test RMSE exceeded **890 USD**, indicating severe overfitting. Feature importance analysis confirmed that **current and lagged gold prices dominate predictions**, reinforcing the importance of autocorrelation in gold price dynamics.

This result highlights a critical limitation of tree-based models: **they cannot extrapolate beyond the range of observed training data**, making them unsuitable for strongly trending financial series.

3.3 Directional Prediction and Trading Performance

Classification models performed poorly, with accuracies near **50%**, barely better than random guessing. Simple trading strategies based on model predictions resulted in **negative cumulative returns**, while a buy-and-hold strategy significantly outperformed all model-based approaches.

This demonstrates that **accurate price forecasting does not necessarily imply profitable trading**, especially when directional accuracy is weak.

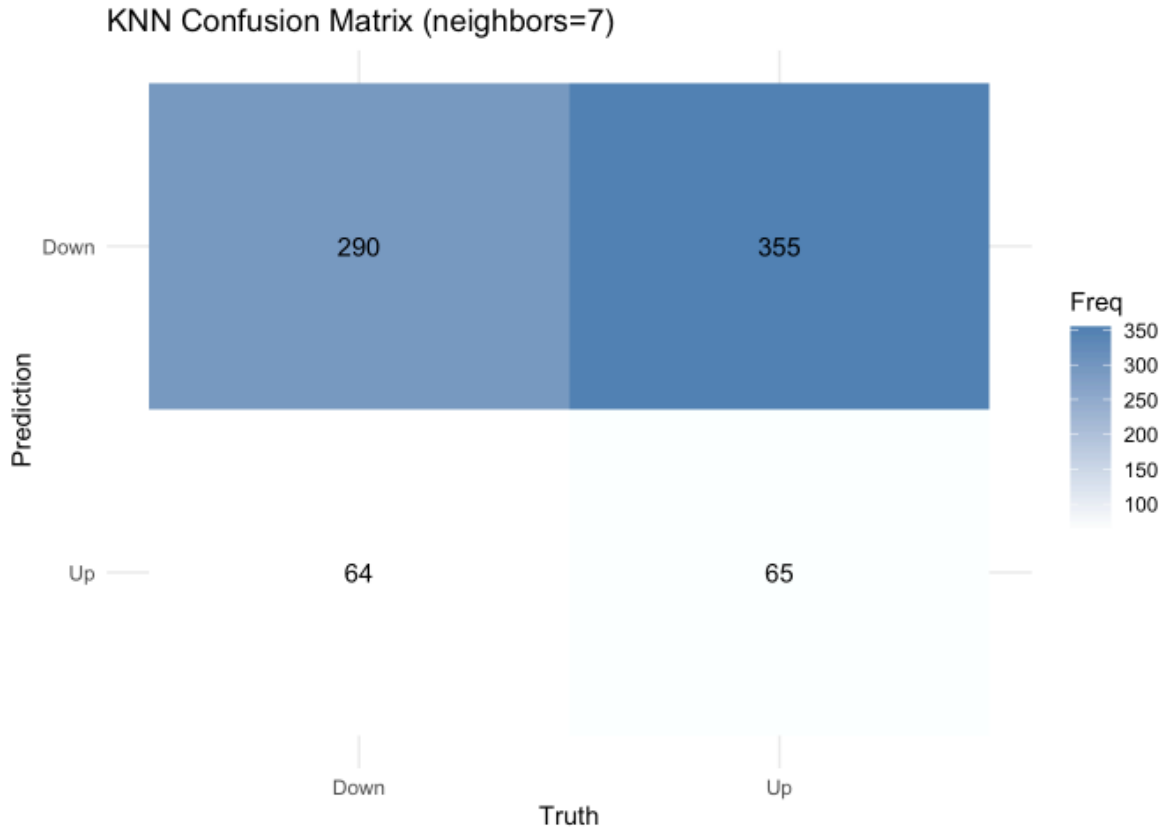

Figure 8: Confusion matrices for classification models

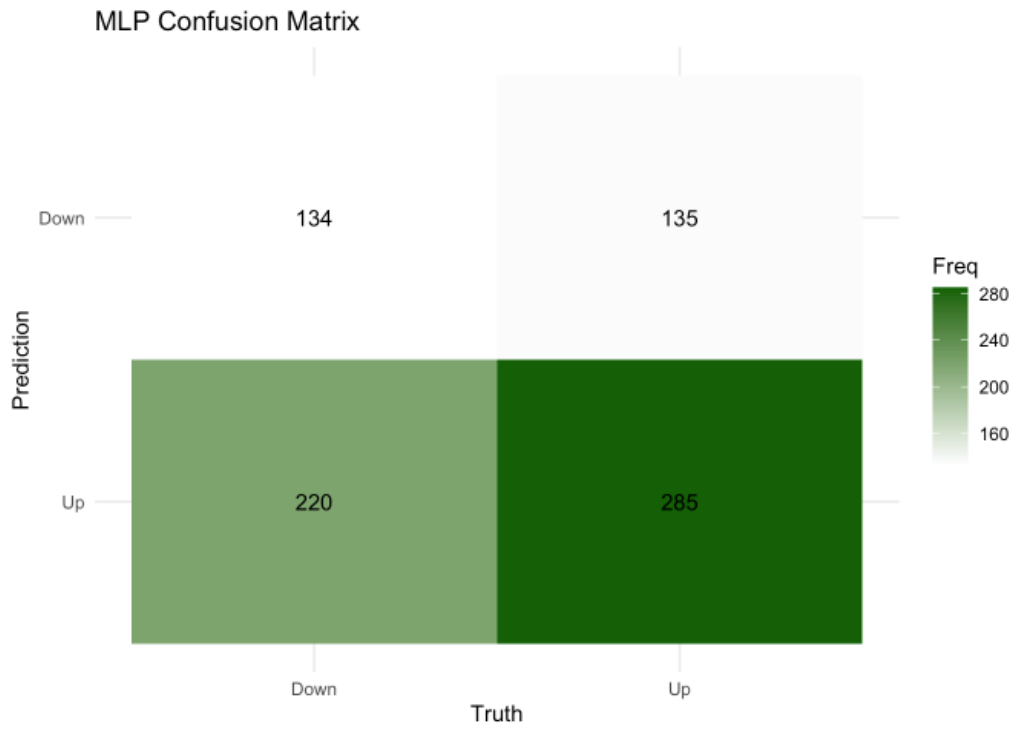

Figure 9: Confusion matrices for MLP

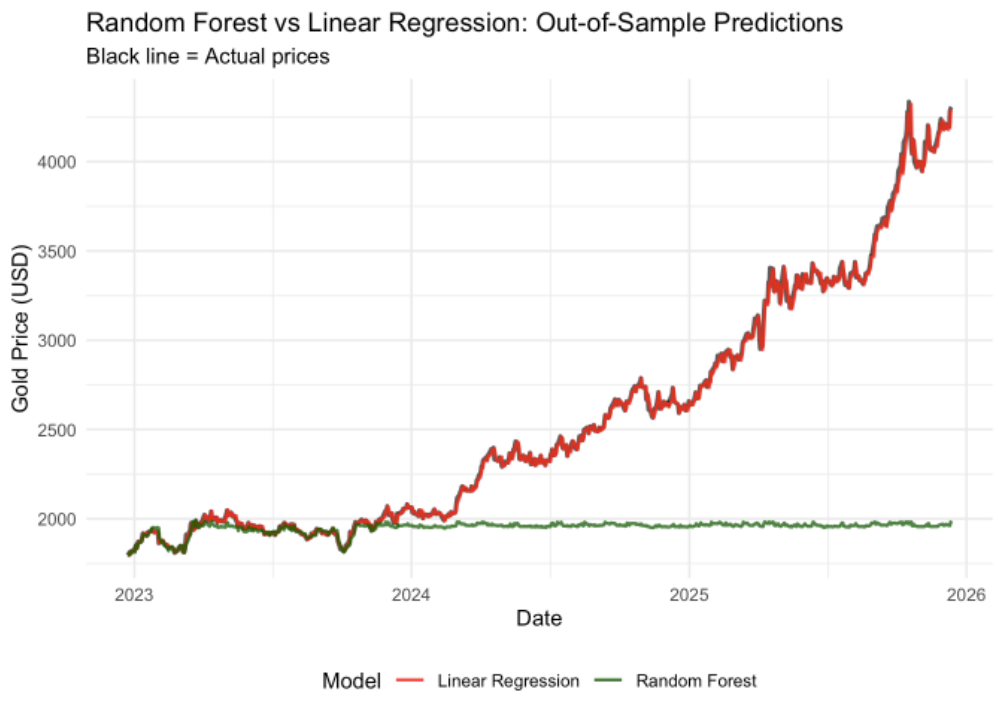

Figure 10: Random Forest vs Linear Regression

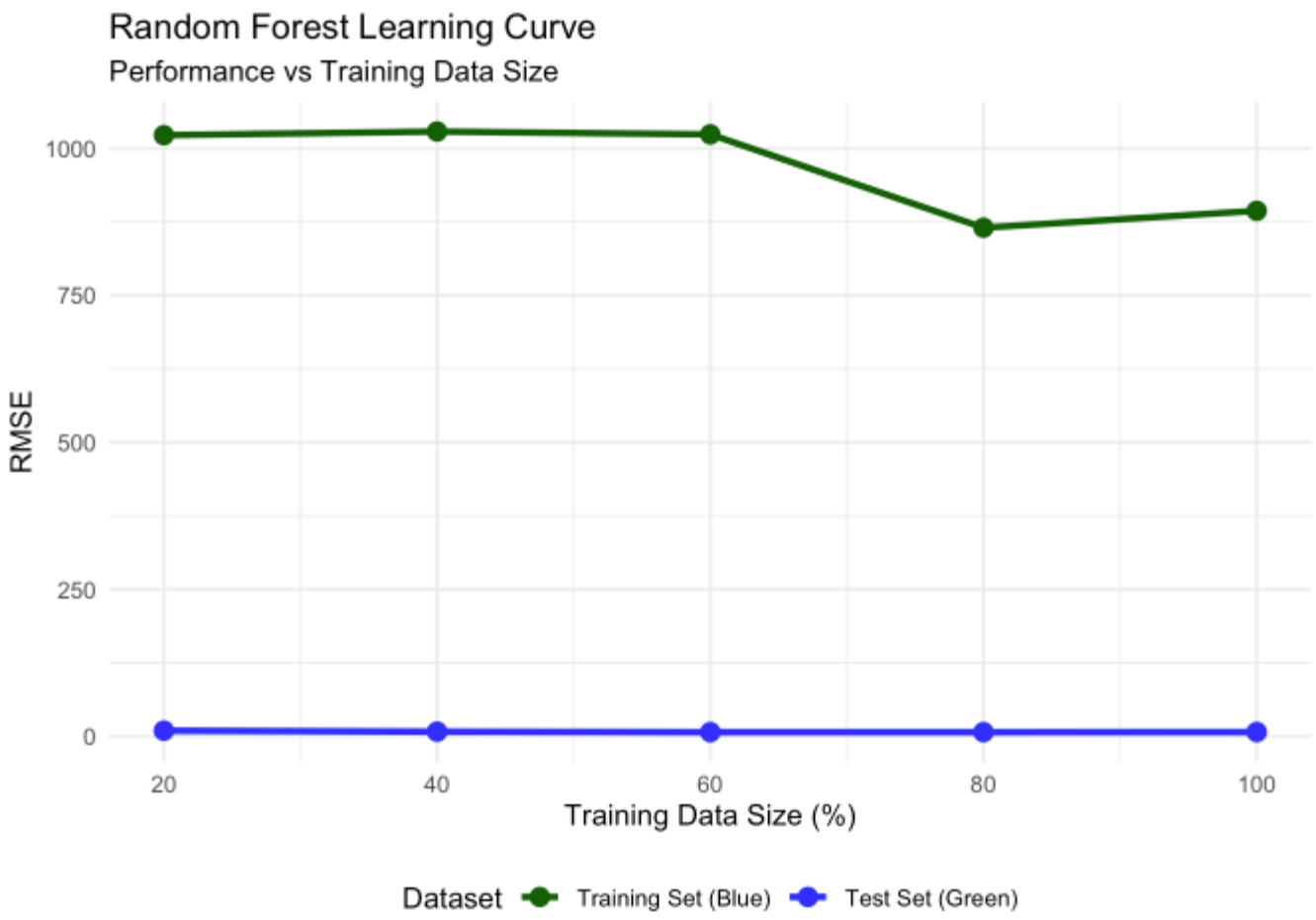

Figure 11: Random Forest Learning Curve

4. Conclusions

This project provides several important insights aligned with the learning objectives of FA-582:

1. **Model simplicity can outperform complexity** in financial forecasting when data exhibits strong trends and autocorrelation.
2. Tree-based machine learning models are ill-suited for extrapolative tasks in time-series data.
3. High predictive accuracy does not guarantee profitable trading strategies.
4. Rigorous out-of-sample testing is essential to avoid misleading conclusions driven by overfitting.

Overall, the results emphasize the importance of **understanding data structure and model assumptions**, rather than blindly applying advanced machine learning techniques.

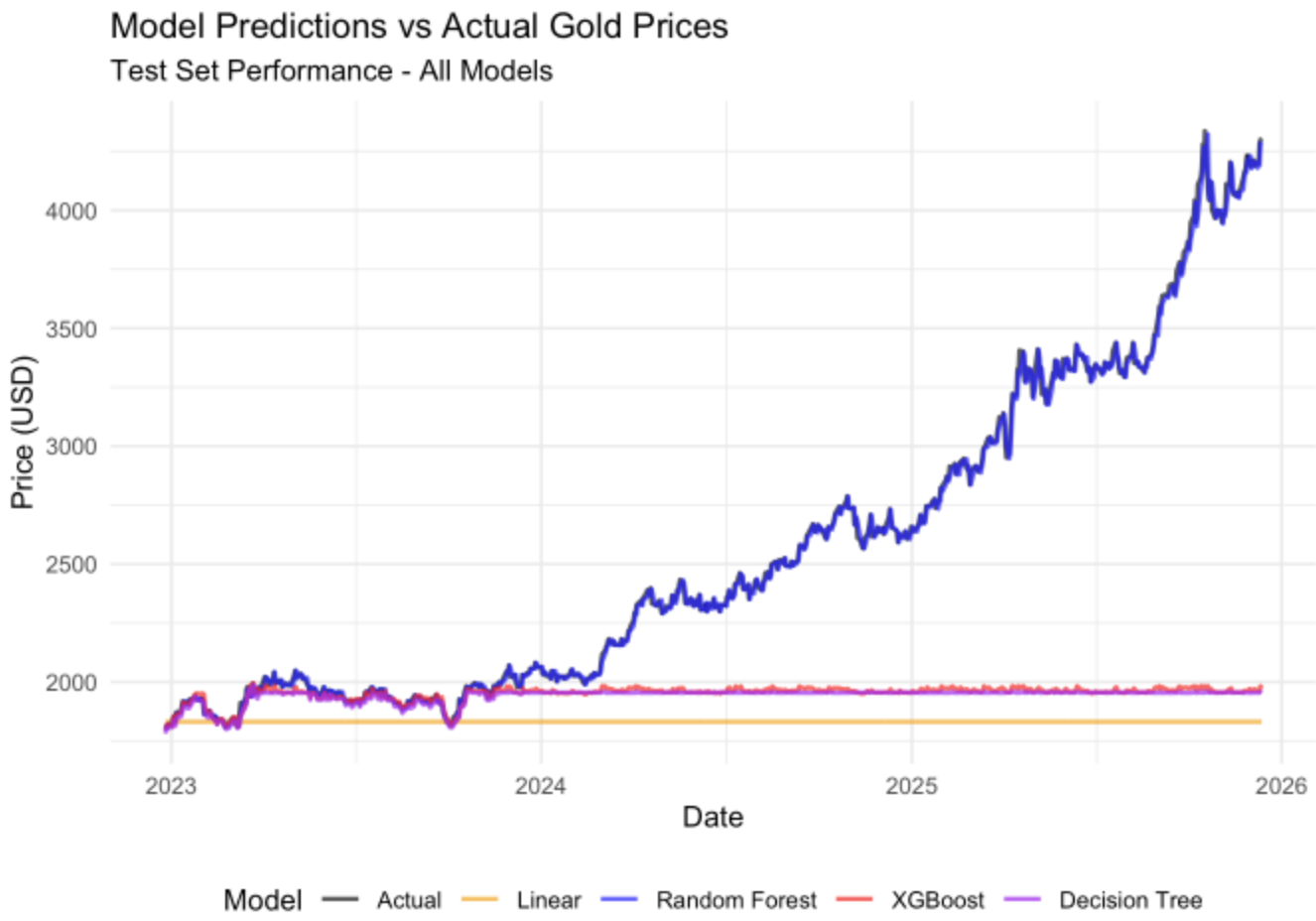

Figure 12: Model Performance vs Actual Gold Prices

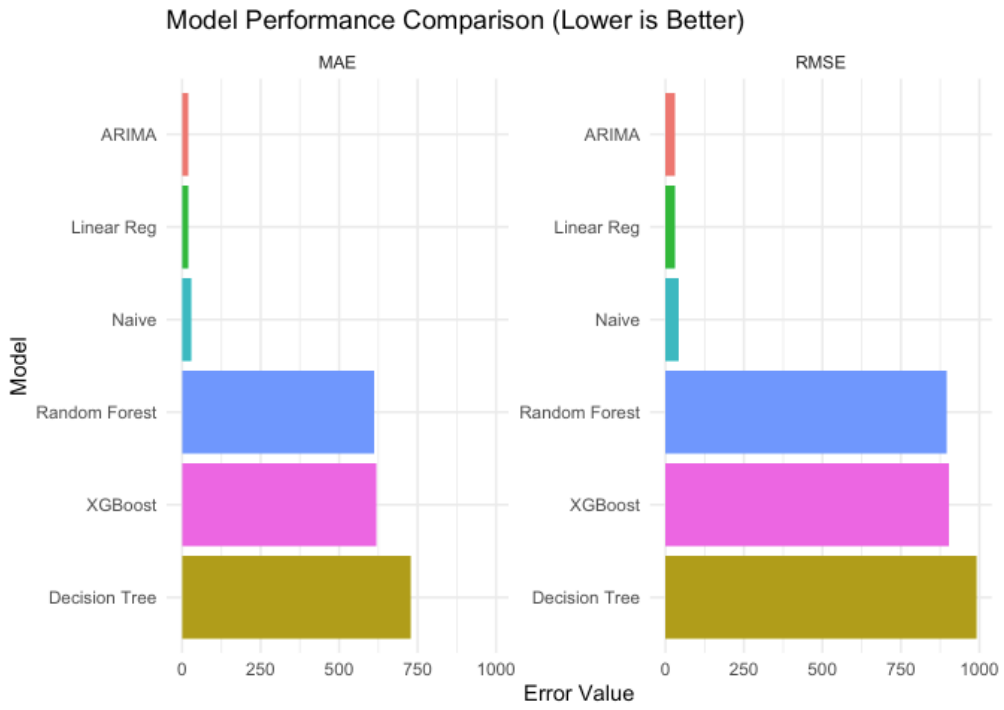

Figure 13: Model Performance Comparison

5. References

- Tsay, R. S. (2010). *Analysis of Financial Time Series*. Wiley.
- Hastie, T., Tibshirani, R., & Friedman, J. (2017). *The Elements of Statistical Learning*. Springer.
- Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice*. OTexts.
- Course lecture notes and project rubric for FA-582

6. Team Members Contribution

Parth Gadekar:

Led the overall project design, data preprocessing pipeline, feature engineering, and regression modeling. Conducted in-depth model diagnostics, overfitting analysis, and performance evaluation. Played a primary role in interpreting results and drafting the final report.

Ankit Dhandharia:

Co-led data acquisition, exploratory data analysis, and implementation of baseline and time-series models (Naïve and ARIMA). Contributed significantly to model comparison, visualization, and result interpretation.

Priti Nagdive:

Assisted with classification models, confusion matrix analysis, and documentation of model performance. Contributed to report formatting and validation of experimental results.

Vedant Ghugare:

Supported exploratory analysis, clustering experiments, and backtesting evaluation. Assisted in preparing visualizations and reviewing the final report for clarity and consistency.